

FN6802 Probability and Statistics

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1. Probability (Part I)

- 1 Probability Measure
- 2 Random Variable and Distribution
- 3 Expectation
- 4 Conditional Expectation

Reading: Rice Ch1-4; Casella and Berger Ch1-2, 4.1-4.3

Further Reading: Roussas Ch1-2, 3.1, 4.1-4.2, 7.1-7.2

Probability Measure

- **Sample space:** Ω is the set of all possible outcomes in a random phenomenon.

for a coin $\Omega = \{h, t\}$

Example 1.1: Throw a coin 3 times,

$$\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}$$

8 outcomes

Example 1.2: Number of jobs in a print queue,

$$\Omega = \{0, 1, 2, \dots\}$$

Example 1.3: Length of time between successive earthquakes,

$$\Omega = \{t | t \geq 0\}$$

Question: What are the differences between Ω 's in these examples?

$$\Omega = \{ \underset{\checkmark}{hhh}, \underline{hht}, \underline{ht h}, \underline{thh}, \underline{htt}, \underline{t h t}, \underline{t t h}, \underline{t t t} \}$$

- **Event:** A particular subset of Ω .

Example 1.4: Let A be the event that total number of heads equals 2 when tossing a coin 3 times. $A = \{hht, hth, thh\}$

Example 1.5: Let A be the event that fewer than 5 jobs in the print queue. $A = \{0, 1, 2, 3, 4\}$ $P(A)$

- **Probability measure:** A function P from subsets of Ω to the real numbers that satisfies the following axioms:

- $$\left\{ \begin{array}{l} \textcircled{1} P(\Omega) = \underline{1}. \\ \textcircled{2} \text{ If } \underline{A} \subset \Omega, \text{ then } \underline{P(A)} \geq 0. \quad 0 \leq P(A) \leq 1 \\ \textcircled{3} \text{ If } \underline{A_1} \text{ and } \underline{A_2} \text{ are disjoint, then} \end{array} \right.$$

$$\underline{P(A_1 \cup A_2) = P(A_1) + P(A_2)}.$$

More generally, if $\underline{A_1, A_2, \dots}$ are mutually disjoint, then

$$\underline{P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)}.$$

- **Example 1.6:** Suppose the coin is fair. For every outcome $\omega \in \Omega$,

$$P(\omega) = 1/8$$

$$P(\Omega) = 1 = P\left(\bigcup_{i=1}^8 \omega_i\right) = P(\Omega) = 1 \quad \Omega = \{ \overset{\omega_1}{hhh}, \overset{\omega_2}{hht}, \overset{\omega_3}{hth}, \overset{\omega_4}{thh}, \overset{\omega_5}{htt}, \overset{\omega_6}{t h t}, \overset{\omega_7}{t t h}, \overset{\omega_8}{t t t} \}$$

- **Conditional probability:** Let A and B be two events with $P(B) > 0$. The conditional probability of A given B is defined to be

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

$$P(A \cap B) = P(A|B)P(B)$$

Example 1.7: Suppose that the first throw is h . What is the probability that we can get exact two h 's in the three trials?

$$\Omega = \{hhh, hht, hth, thh, htt, tth, ttt\}$$

$$P(A) = \frac{3}{8}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/8}{4/8} = \frac{1}{2} = \frac{\{hhh, hht, hth, htt\}}{\{hhh, hht, hth, htt, thh, tth, ttt\}} = \frac{\{hht, hth, thh\}}{\{hht, hth, thh\}}$$

$B = \text{1st toss gives } h$
 $A = \text{exact 2 } h\text{'s}$

- **Multiplication law:** Let A and B be events and assume $P(B) > 0$. Then

$$P(A \cap B) = P(A|B)P(B).$$

Example 1.8: Three prisoners A, B and C are on death row. The governor decides to pardon one of them and chooses the prisoner to pardon at random. He informs the warden of this choice but requests that the name be kept secret for a few days. A tries to get the warden to tell him who had been pardoned. The warden refuses. A then asks which of B or C will be executed. The warden thinks for a while, and tells A that B is to be executed. A then thinks that his chance of being pardoned has risen to $\frac{1}{2}$. Is he right? *No.*

Let X = the prisoner to be pardoned. $P(X=A) = P(X=B) = P(X=C) = \frac{1}{3}$

Let Y = the prisoner that warden says to be executed.

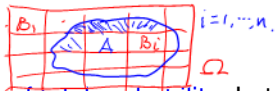
$$P(X=A | Y=B)$$

$$= \frac{P(X=A, Y=B)}{P(Y=B)} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}$$

$$= P(Y=B, X=A) + P(Y=B, X=B) + P(Y=B, X=C)$$

$$= \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

Prob	$\Omega_X = \{A, B, C\}$	Y	$\Omega_Y = \{B, C\}$
$\frac{1}{3}$	A	B	$P(Y=B) + P(Y=C) = 1/3$
$\frac{1}{3}$	B	C	$P(Y=B) = P(Y=C) = 1/6$
$\frac{1}{3}$	C	B	$P(Y=C) = P(X=B) = 1/3$
		B	$P(Y=B) = \frac{1}{3}$



$$\bigcup_{i=1}^n (B_i \cap A) = A, \quad P(A) = P\left(\bigcup_{i=1}^n (B_i \cap A)\right) = \sum_{i=1}^n P(B_i \cap A) \quad \checkmark$$

- **Law of total probability:** Let $B_1, B_2, \dots, B_n$ be events such that $\bigcup_{i=1}^n B_i = \Omega$ and $B_i \cap B_j = \emptyset$ for $i \neq j$, with $P(B_i) > 0$ for all i . Then for any event A

$$P(B_i \cap A) = P(A|B_i)P(B_i)$$

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i) \quad \checkmark$$

- **Bayes' Rule:** Let A and $B_1, B_2, \dots, B_n$ be events where B_i are disjoint, $\bigcup_{i=1}^n B_i = \Omega$ and $P(B_i) > 0$ for all i . Then

$$\frac{P(A|B_j) \cdot P(B_j)}{P(A)} = P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

- **Independence:** Two events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

A group of events $A_1, A_2, \dots, A_n$ are said to be mutually independent if for any subgroup $A_{i_1}, \dots, A_{i_m}$,

$$P\left(\bigcap_{j=1}^m A_{i_j}\right) = \prod_{j=1}^m P(A_{i_j}).$$

Example for Bayes' Rule: A computer produced by a company was found to be defective (D). There are three factories (A, B, C) where such computers are manufactured. A Quality Control Manager (QCM) was responsible for investigating the source of the defect. Below is what the QCM knew about the company's computer production and the possible source of defects:

marginal

Factory	proportion of total production	Prob of defective products
A	$0.35 = P(A)$	$1.5\% = P(D A)$
B	$0.35 = P(B)$	$1\% = P(D B)$
C	$0.30 = P(C)$	$2\% = P(D C)$

The QCM: if a randomly selected computer was defective, what was the probability that the computer was manufactured in factory C?

$$P(C|D) = \frac{P(C \cap D)}{P(D)} = \frac{P(D|C) \cdot P(C)}{P(D|A)P(A) + \dots + P(D|C)P(C)} = \frac{2\% \cdot 30\%}{1.5\% \times 35\% + 1\% \times 35\% + 2\% \times 30\%}$$

$= \frac{P(D \cap A) + P(D \cap B) + P(D \cap C)}{P(D)}$

$$= P(D|A)P(A) + \dots + P(D|C)P(C).$$

Random Variable and Distribution

$$\Omega \xrightarrow{f} \mathbb{R} \quad f(\omega) \in \mathbb{R}$$

- Random variable: is a function from Ω to the real numbers

Example 1.9: X_1 = the total number of heads $X_1 \triangleq f(\omega) \in \mathbb{R} = \{0, 1, 2, 3\}$.

X_2 = the number of heads on the first toss

X_3 = the number of heads minus the number of tails

$$\Omega = \{ \underline{hhh} \dots \underline{ttt} \},$$

$$X_2 \in \mathbb{R} = \{0, 1\}.$$

$$X_3 \in \mathbb{R} = \{3, 1, -1, -3\}.$$

X_4 = duration of 2 successive investments

$$\mathbb{R} = \{x \mid x > 0\}.$$

X_5 = profit of a company, $X_5 \in \mathbb{R}$.

Question: Why statisticians need random variables? why they map to real line? observation = data.

Distribution

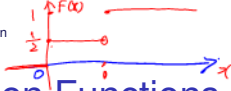


Question: A random variable has a sample space on real line. Does it bring some special ways to describe its probability measure?

	Discrete	Continuous
Uni- variate r.v.	pmf cdf mgf	pdf cdf mgf
Multi- variate r.v.	joint pmf joint cdf joint mgf	joint pdf joint cdf joint mgf

pmf: probability mass function, pdf: probability density function,
cdf: cumulative distribution function,
mgf: moment generating distribution.

- A **discrete random variable** can take on only a finite or at most a countably infinite number of values. A **continuous random variable** can take on a continuum of values.



$$F(x) = P(X \leq x) \leq F(x^*) = P(X \leq x^*) \\ x^* > x$$



Distribution Functions

to discrete

$$X = \begin{cases} 1 & P = 1/2 \\ 0 & P = 1/2 \end{cases}$$

$$F(x) = P(X \leq x) = \begin{cases} 1 & x \geq 1 \\ 1/2 & 0 \leq x < 1 \\ 0 & x < 0 \end{cases}$$

- A function F is the **Cumulative distribution function (cdf)** of a random variable X if $F(x) = P(X \leq x)$, $x \in \mathbb{R}$.

$P(0.5 \leq X < 1.5) = \sum P(x) = P(1) = 1/2$

- A function $p(x)$ is a **Probability mass function (pmf)** if and only if $p(x) \geq 0$ for all x and $\sum_{-\infty < x < \infty} p(x) = 1$. For a discrete random variable X with pmf $p(x)$, $P(X = x) = p(x)$ and $P(X \in A) = \sum_A p(x)$.

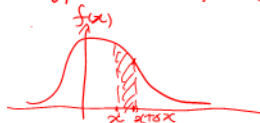
- A function $f(x)$ is a **Probability density function (pdf)** if and only if $f(x) \geq 0$ for all x and $\int_{-\infty}^{\infty} f(x) dx = 1$. For a continuous random variable X with pdf $f(x)$, $P(X \in A) = \int_A f(x) dx$.

$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$, $f(x) = \frac{d}{dx} F(x)$

$\lim_{\Delta x \rightarrow 0} \frac{F(x+\Delta x) - F(x)}{\Delta x} = P(X \in (1, 2) | X \sim N(0, 1)) = \int_1^2 f(x) dx$, $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

Note. $\forall x$ st $f(x) > 0$, $P(X = x) = 0$.

Question: How to interpret $f(x)$?



Properties of cdf

If $F(x)$ is a cdf of some r.v. X then the following properties hold.

- ➊ $0 \leq F(x) \leq 1$.
- ➋ $F(x)$ is nondecreasing.
- ➌ For any $x \in \mathbb{R}$, $F(x)$ is continuous from the right, i.e.,

$$\lim_{t \downarrow x} F(t) = F(x).$$

- ➍ $\lim_{x \rightarrow \infty} F(x) = 1, \lim_{x \rightarrow -\infty} F(x) = 0$.
- ➎ $P(X > x) = 1 - F(x)$ and $P(a < X < b) = F(b) - F(a)$.
- ➏ For any $x \in \mathbb{R}$, $F(x)$ has left limit.
- ➐ There are at most countably many discontinuity points of $F(x)$.

Conversely, if a function $F(x)$ satisfies properties 2, 3, 4 then $F(x)$ is a cdf.

Joint Distribution

- Question:** Why we need joint distribution to describe multivariate r.v.'s?

Example 1.10 (cont. Example 1.9):

$$\Omega = \{hhh, hht, hth, thh, \underline{htt}, \underline{tth}, \underline{ttt}\}$$

marginal dist of X_1 .
↓
pmf of X_1

X_2 : no. of heads on 1st toss		X_1 : total no. of heads			
		0 ($\frac{1}{8}$)	1 ($\frac{3}{8}$)	2 ($\frac{3}{8}$)	3 ($\frac{1}{8}$)
pmf of X_2 marginal dist of X_2	0 ($\frac{1}{2}$)	$P(X_1=0, X_2=0) = \frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$P(X_1=3, X_2=0) = 0$
	1 ($\frac{1}{2}$)	$P(X_1=0, X_2=1) = 0$	$\frac{1}{8}$	$\frac{2}{8}$	$P(X_1=3, X_2=1) = \frac{1}{8}$

Joint pmf of (X_1, X_2)

- Question:** When we know the joint cdf, we can obtain every marginal cdf. Is the reverse statement true?

Joint/Marginal Distribution Functions

- The **joint cdf** of $X_1, \dots, X_n$ is $F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$ for $x_1, \dots, x_n \in \mathbb{R}$.
- The **marginal cdf** of X_1 is

$$F_{X_1}(x_1) = P(X_1 \leq x_1) = \lim_{x_2, \dots, x_n \rightarrow \infty} F(x_1, \dots, x_n).$$

- Discrete case: **marginal pmf** $p_{X_1}(x) = F_{X_1}(x) - F_{X_1}(x-)$.
- Continuous case: **marginal pdf** $f_{X_1}(x) = \frac{d}{dx} F_{X_1}(x)$.

- Discrete multivariate case: $p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$
 $\Rightarrow$ joint pmf of $X_1, \dots, X_n$

$$P((X_1, \dots, X_n) \in A) = \sum_{(x_1, \dots, x_n) \in A} p(x_1, \dots, x_n)$$

$$F(x_1, \dots, x_n) = \sum_{t_1 \leq x_1, \dots, t_n \leq x_n} p(t_1, \dots, t_n)$$

$$p_{X_1}(x_1) = P(X_1 = x_1) = \sum_{-\infty < t_2 < \infty, \dots, -\infty < t_n < \infty} p(x_1, t_2, \dots, t_n)$$

- Continuous multivariate case: $f(x_1, \dots, x_n) = \frac{\partial^n}{\partial x_1 \dots \partial x_n} F(x_1, \dots, x_n)$
 $\Rightarrow$ joint pdf of $X_1, \dots, X_n$

$$P((X_1, \dots, X_n) \in A) = \int \dots \int_A f(x_1, \dots, x_n) dx_1 \dots dx_n$$

$$F(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f(t_1, \dots, t_n) dt_n \dots dt_1$$

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, t_2, \dots, t_n) dt_2 \dots dt_n$$

Independent Random Variables

- Random variables $X_1, \dots, X_n$ are said to be **independent** if their joint cdf factors into the product of their marginal cdf's

$$F(x_1, \dots, x_n) = F_{X_1}(x_1) \dots F_{X_n}(x_n)$$

for all $x_1, \dots, x_n$.

- 1 For discrete case:

$$F(x_1, \dots, x_n) = F_{X_1}(x_1) \dots F_{X_n}(x_n) \Leftrightarrow p(x_1, \dots, x_n) = p_{X_1}(x_1) \dots p_{X_n}(x_n)$$

For continuous case:

$$F(x_1, \dots, x_n) = F_{X_1}(x_1) \dots F_{X_n}(x_n) \Leftrightarrow f(x_1, \dots, x_n) = f_{X_1}(x_1) \dots f_{X_n}(x_n)$$

- 2 X and Y are independent

$$\Leftrightarrow P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

i.e. the events $\{X \in A\}$ and $\{Y \in B\}$ are independent.

- ⑧ X and Y are independent $\Rightarrow Z = \underline{g(X)}$ and $W = \underline{h(Y)}$ are independent.

Proof. $\underline{F(z, w)} = P(Z \leq z, W \leq w)$ $A = \{x : \underline{g(x)} \leq z\}$,
 $B = \{y : \underline{h(y)} \leq w\}$.

$$= P(g(X) \leq z, h(Y) \leq w)$$

$$= P(X \in A, Y \in B)$$

$$\stackrel{\text{independent}}{=} P(X \in A) \cdot P(Y \in B)$$

$$= P(g(X) \leq z) P(h(Y) \leq w)$$

$$= P(Z \leq z) \cdot P(W \leq w)$$

$$= F_Z(z) \cdot F_W(w) \Rightarrow Z \& W \text{ indep.}$$

- ⑨ Marginal distributions of $X_1, \dots, X_n$ + independence $\Leftrightarrow$ joint distribution of $X_1, \dots, X_n$.

" $\Leftarrow$ " $f(x_1, x_2) \Rightarrow f_{X_1}(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2$ marginal dist.

" $\Rightarrow$ " $f_{X_1}(x_1) \cdot f_{X_2}(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2$

Conditional Distribution

- X and Y are discrete random variables with joint pmf $p_{XY}(x, y)$, the conditional pmf of Y given X is

$$\underline{p_{Y|X}(y|x)} = \underline{P(Y = y|X = x)} = \frac{P(X=x, Y=y)}{P(X=x)} \quad P(X=x) > 0.$$

$$= \frac{p_{XY}(x, y)}{p_X(x)}$$

if $p_X(x) > 0$. The probability is defined to be zero if $p_X(x) = 0$.

Example (cont. Ex 1.10): Conditional dist of X_2 given $X_1 = 1$.

$$\underline{p_{X_2|X_1}(0|1)} = \frac{\frac{2}{8}}{\frac{3}{8}} = \frac{p(X_1=1, X_2=0)}{p(X_1=1)} \quad \text{and} \quad \underline{p_{X_2|X_1}(1|1)} = \frac{p(X_1=1, X_2=1)}{p(X_1=1)} = \frac{1/8}{3/8} = \frac{1}{3}$$

- X and Y are continuous random variables with joint pdf $f_{XY}(x, y)$, the conditional pdf of Y given X is

$$\underline{f_{Y|X}(y|x)} = \frac{f_{XY}(x, y)}{f_X(x)}, \quad y \in \mathbb{R}, \quad f_X(x) > 0.$$

if $0 < f_X(x) < \infty$ and 0 otherwise.

Some remarks:

① For each fixed x , $p_{Y|X}(y|x)$ is a pmf for y and $f_{Y|X}(y|x)$ is a pdf for y .

② $p_{XY}(x,y) = p_{Y|X}(y|x)p_X(x)$ and $f_{XY}(x,y) = f_{Y|X}(y|x)f_X(x)$

$\Rightarrow$ Multiplication Law.

③ $p_Y(y) = \sum_x p_{Y|X}(y|x)p_X(x)$ and $f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y|x)f_X(x)dx$

$\Rightarrow$ Law of total probability.

④ X, Y are independent $\Leftrightarrow p_{Y|X}(y|x) = p_Y(y)$ or $f_{Y|X}(y|x) = f_Y(y)$.

Functions of random variables

Question: For given r.v.'s $X_1, \dots, X_n$, how to derive the distributions of their transformations?

- ① **Method of events:** Let $\mathbf{X} = (X_1, \dots, X_n)$ be random variables and $\mathbf{Y} = g(\mathbf{X})$. Then the distribution of $\mathbf{Y}$ is determined as follow: for any event B defined by $\mathbf{Y}$, $P(\mathbf{Y} \in B) = P(\mathbf{X} \in A)$, where $A = g^{-1}(B)$.

Example 1.11 (univariate discrete r.v.): Let X be a discrete r.v. taking the values $x_i, i = 1, 2, \dots$, and $Y = g(X)$. Then Y is also a discrete r.v. taking the values $y_j, j = 1, 2, \dots$. To determine the pmf of Y , by taking $B = \{y_j\}$, we have

$$A = \{x_i : g(x_i) = y_j\}$$

and hence

$$\begin{aligned} p_Y(y_j) &= P(\{y_j\}) = P(A) = \sum_{x_i \in A} p_X(x_i) \\ &= P(Y = y_j) = P(g(X) = y_j) \\ &= P(X \in A) \end{aligned}$$

$$\begin{aligned} &\{Y = y_j\} \\ &= \{g(X) = y_j\} \\ &= \{x_i : g(x_i) = y_j\} \end{aligned}$$

$$P(X \in A) = \sum_{x \in A} p(x)$$

$$P((X, Y) \in A) = \sum_{(x, y) \in A} p(x, y)$$

Example 1.12 (sum of two discrete r.v.'s): X and Y are random variables with joint pmf $p(x, y)$. Find the distribution of $\underline{Z = X + Y}$.
(Exercise 1: Find the distribution of $\underline{Z = Y - X}$)

Solution.

pmf of $\underline{Z}$: $\underline{p_z(z)} = P(Z=z)$ $\underline{A = \{(x, y) : x+y=z\}}$

$$= P(X+Y=z)$$

$$= \sum_{(x,y) \in A} p(x, y)$$

Suppose $X = X_1$ (total no. of heads), $Y = X_2$ (no of head in 1st toss).
 $\{0, 1, 2, 3, \}$ $\{0, 1\}$.

$$\underline{Z = X - Y}$$

$Z = X+Y$, outcome: $\{0, 1, 2, 3, 4\}$. $\Rightarrow Z$

pmf $p_z(0) = P(X+Y=0) = P(X=0, Y=0) = \frac{1}{8}$

$$p_z(1) = P(X+Y=1) = P(X=0, Y=1) + P(X=1, Y=0) = 0 + \frac{2}{8} = \frac{2}{8}$$

$$p_z(2) = \frac{2}{8}, \quad p_z(3) = \frac{2}{8}$$

$$p_z(4) = P(X+Y=4) = P(X=3, Y=1) = \frac{1}{8}$$

$$\text{CDF of } Y: F_Y(y) = P(Y \leq y) = P(g(X) \leq y) \stackrel{\substack{\text{monotone} \\ \text{if } g \text{ increasing}}}{=} \underline{P(X \leq g^{-1}(y))} = F_X(g^{-1}(y))$$

② **Method of cdf** (a special case of method 1): Let Y be a function of r.v.'s $X_1, \dots, X_n$.

- i. Find the region $Y \leq y$ in the $(x_1, \dots, x_n)$ space.
- ii. Find $F_Y(y) = P(Y \leq y)$ by summing the joint pmf or integrating the joint pdf of $X_1, \dots, X_n$ over the region $Y \leq y$.
- iii. (For continuous case) Find the pdf of Y by differentiating $F_Y(y)$, i.e.,
 $f_Y(y) = \frac{d}{dy} F_Y(y)$.

Note. It can be generalized to multivariate $\mathbf{Y} = (Y_1, \dots, Y_m)$.

$$P((X, Y) \in A) = \iint_A f(x, y) dx dy$$

Example 1.13 (sum of two continuous r.v.'s): X and Y are random variables with joint pdf $f(x, y)$. Find the distribution of $Z = X + Y$.

(Exercise 2: Find the distribution of $Z = Y - X$)

Solution. cdf of Z , $F(z) = P(Z \leq z)$ let $A = \{(x, y) : x + y \leq z\}$
 $= P(X + Y \leq z)$
 $= \iint_A f(x, y) dx dy$ $x \leq z - y$ $-\infty < y \leq z - x$

If X, Y indep, $f(x, y) = f_X(x) \cdot f_Y(y)$

$$f(z) = \frac{dF(z)}{dz}$$

$$= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

$$F(z) = \int_A f_X(x) \cdot f_Y(y) dx dy = \int_{-\infty}^{\infty} f_X(x) \left[\int_{-\infty}^{z-x} f_Y(y) dy \right] dx = \int_{-\infty}^{\infty} f_X(x) F_Y(z-x) dx$$

Suppose $X, Y \sim \text{Exp}(\lambda)$, pdf $f(x) = \lambda e^{-\lambda x}$ for $x > 0$, $y > 0$, $0 < x, y \leq z$.
 $F(x) = 1 - e^{-\lambda x}$

$$\text{cdf: } F(z) = \int_0^z \lambda e^{-\lambda x} (1 - e^{-\lambda(z-x)}) dx$$

$$\text{pdf of } Z: f_Z(z) = \frac{dF(z)}{dz} = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx = \lambda^2 e^{-\lambda z} \cdot z$$

(Exercise 3: Find the distribution of product of 2 r.v.'s $Z = XY$)

E.g. $Y = X - 2$. $X \sim \text{pdf } f(x)$. $f_Y(y) = f(y+2)$

$\approx f(y+2)$ correct?

$X = Y + 2$

Method of pdf (a special case of method 2 for **continuous** r.v. and **differentiable, 1-to-1 transformations**):

Univariate case: Let X be a continuous r.v. with pdf $f_X(x)$. Let $Y = g(X)$, where g is differentiable, strictly monotone. Then,

$\underbrace{f_X(x)}_{f_X(x)} \xrightarrow{g(\cdot)} \underbrace{f_Y(y)}_{f_Y(y)=?}$ $Y = g(X) \Rightarrow X = g^{-1}(Y)$ Jacobian factor.

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|$$

for y s.t. $y = g(x)$ for some x , and $f_Y(y) = 0$ otherwise.

Proof. $F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = \begin{cases} P(X \leq g^{-1}(y)), & \text{if } g \text{ is increasing} \\ P(X \geq g^{-1}(y)), & \text{if } g \text{ is decreasing} \end{cases}$

$$= \begin{cases} F_X(g^{-1}(y)), & \text{increasing} \\ 1 - F_X(g^{-1}(y)), & \text{decreasing} \end{cases}$$

pdf of z : $f_Z(z) = \frac{dF_Y(y)}{dy} = \begin{cases} f_X(g^{-1}(y)) \cdot \frac{dg^{-1}(y)}{dy} \\ -f_X(g^{-1}(y)) \cdot \frac{dg^{-1}(y)}{dy} \end{cases} = f_X(g^{-1}(y)) \cdot \left| \frac{dg^{-1}(y)}{dy} \right|$

magnification factor.

Question. What is the role of $|dg^{-1}(y)/dy|$? How to interpret it?

Multivariate case: $\mathbf{X} = (X_1, \dots, X_n)$ multivariate continuous,
 $\mathbf{Y} = (Y_1, \dots, Y_n) = \mathbf{g}(\mathbf{X})$. $\mathbf{g}$ is one-to-one so that its inverse exists
 and is denoted by

$$\mathbf{x} = \mathbf{g}^{-1}(\mathbf{y}) = \mathbf{w}(\mathbf{y}) = (w_1(\mathbf{y}), \dots, w_n(\mathbf{y})).$$

Handwritten notes: $\frac{\partial f(\vec{x})}{\partial \vec{x}}$ on the left; $\frac{\partial \mathbf{g}^{-1}(\vec{y})}{\partial \vec{y}}$ on the right.

Assume $\mathbf{w}$ have continuous partial derivatives, and let

$$J = \begin{vmatrix} \frac{\partial w_1(\mathbf{y})}{\partial y_1} & \dots & \frac{\partial w_1(\mathbf{y})}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial w_n(\mathbf{y})}{\partial y_1} & \dots & \frac{\partial w_n(\mathbf{y})}{\partial y_n} \end{vmatrix}$$

determinant of a matrix

Then

$$f_Y(\mathbf{y}) = f_X(\mathbf{g}^{-1}(\mathbf{y})) |J|,$$

Handwritten notes: "abs." with an arrow pointing to |J|; "marginal $f_{Y_i}(y_i)$ " with an arrow pointing to the equation.

$$= \int \int f_X(y_1, \dots, y_n) dy_2 \dots dy_n$$

for $\mathbf{y}$ s.t. $\mathbf{y} = \mathbf{g}(\mathbf{x})$ for some $\mathbf{x}$, and $f_Y(\mathbf{y}) = 0$ otherwise.

$$\int f_Y(y_1, y_2) dy_2 \rightarrow f_{Y_1}(y_1)$$

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \rightarrow \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} X_2/X_1 \\ X_1 \end{pmatrix}$$

$$\begin{aligned} Y_1 &= X_2/X_1 \Rightarrow X_2 = X_1 Y_1 = Y_1 Y_2 \\ Y_2 &= X_1 \Rightarrow X_1 = Y_2 \end{aligned}$$

Example 1.14: X_1 and X_2 are random variables with joint pdf $f_{X_1 X_2}(x_1, x_2)$. Find the distribution of $Y_1 = X_2/X_1$.

(Exercise 4: Find the distribution of $Y_1 = X_1 X_2$)

Solution.

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \Rightarrow \begin{pmatrix} Y_2 \\ Y_1 Y_2 \end{pmatrix} \begin{matrix} w_1(y) \\ w_2(y) \end{matrix}$$

$$f_Y(y_1, y_2) = f_X(y_2, y_1 y_2) \cdot |J| = f(y_2, y_1 y_2) \cdot |y_2|$$

$$J = \begin{vmatrix} \frac{\partial w_1(y)}{\partial y_1} & \frac{\partial w_1(y)}{\partial y_2} \\ \frac{\partial w_2(y)}{\partial y_1} & \frac{\partial w_2(y)}{\partial y_2} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ y_2 & y_1 \end{vmatrix} = -y_2$$

$$f_{Y_1}(y_1) = \int_{-\infty}^{\infty} f_Y(y_1, y_2) dy_2 = \int_{-\infty}^{\infty} f(y_2, y_1 y_2) \cdot |y_2| dy_2$$

Expectation

Question: Can we describe the characteristics of distributions by use of some intuitive and meaningful simple values?

- For r.v.'s $X_1, \dots, X_n$, the **expectation** of a univariate r.v.

$Y = g(X_1, \dots, X_n)$ is defined by

$X \sim \text{pdf } f(x) \text{ or pmf } p(x)$

$\checkmark E(X) = \sum_{-\infty < x < \infty} p(x) \cdot x$

$$\begin{aligned}
 E(Y) &= E[g(X_1, \dots, X_n)] = \sum_{-\infty < y < \infty} y p_Y(y) \\
 &= \sum_{-\infty < x_1 < \infty, \dots, -\infty < x_n < \infty} g(x_1, \dots, x_n) p(x_1, \dots, x_n)
 \end{aligned}$$

if $X_1, \dots, X_n$ are discrete, or

$E(X) = \int_{-\infty}^{\infty} x f(x) dx$

$$\begin{aligned}
 E(Y) &= E[g(X_1, \dots, X_n)] = \int_{-\infty}^{\infty} y f_Y(y) dy \\
 &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} g(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \dots dx_n,
 \end{aligned}$$

if Y and $X_1, \dots, X_n$ are continuous.

- For any decision facing uncertain outcomes, we can evaluate it by the expectation of the decision.

Example 1.15: Suppose you are offered a game: You pay \$3 for a ticket to toss a pair of dice, if it is a (6,6), you win \$100, otherwise you win nothing. Decide if you will play the game or not.

Solution. Let $X = \begin{cases} 100-3, & P(6,6) = \frac{1}{36} \checkmark \\ -3, & P(\text{others}) = \frac{35}{36} \checkmark \end{cases}$

$$E(X) = 97 * \frac{1}{36} + (-3) * \frac{35}{36} = \frac{97 - 105}{36} = -\frac{8}{36} = -\frac{2}{9}$$

Question: If you are offered \$2 a ticket to play the same game, will you decide to play?

$$X = \begin{cases} 98, & p = \frac{1}{36} \\ -2, & p = \frac{35}{36} \end{cases} \quad E(X) = \frac{7}{9}$$

Various Characteristic Measures

- $g(x) = x \Rightarrow E[g(X)] = E(X)$ is called mean of X , denoted by μ_X . *central tendency*
- $g(x) = (x - \mu_X)^2 \Rightarrow E[g(X)] = E[(X - E(X))^2]$ is called variance of X , denoted by $Var(X)$ or σ_X^2 . Its square root σ_X is called standard deviation. *variation or dispersion*
- $g(x, y) = (x - \mu_X)(y - \mu_Y) \Rightarrow E[g(X, Y)] = E[(X - E(X))(Y - E(Y))]$ is called covariance of X and Y , denoted by $Cov(X, Y)$ or σ_{XY} .
- The correlation coefficient of X and Y is defined by $\sigma_{XY} / (\sigma_X \sigma_Y)$, denoted by $Cor(X, Y)$ or ρ_{XY} . X and Y are uncorrelated if $\rho_{XY} = 0$. *$-1 \leq \rho_{XY} \leq 1$*

Note. All of them are constants, NOT random variables!

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x,y) dx dy \stackrel{\text{indep}}{=} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_x(x) f_y(y) dx dy$$

$$= \underbrace{\int_{-\infty}^{\infty} x f_x(x) dx} \cdot \underbrace{\int_{-\infty}^{\infty} y f_y(y) dy}$$

Properties

$$Eg(x) = \sum_{i=1}^{\infty} g(x_i) p(x_i)$$

$$E(ax + by) = aEX + bEY$$

$$E(X^2 - 2X \cdot EX + (EX)^2)$$

$$\bullet \text{ } Var(X) = E[(X - E(X))^2] = E(X^2) - \mu_X^2;$$

$$Cov(X, Y) = E(XY) - \mu_X \mu_Y.$$

$$E(XY) \neq E(X) \cdot E(Y).$$

$$\bullet \text{ } Var(a + bX) = b^2 Var(X), a, b \in R \text{ and } \sigma_{a+bX} = |b| \sigma_X.$$

$$\bullet \text{ } Var(a + \sum_{i=1}^n b_i X_i) = \sum_{i=1}^n b_i^2 Var(X_i) + 2 \sum_{1 \leq i < j \leq n} b_i b_j Cov(X_i, X_j);$$

$$Cov(a + \sum_{i=1}^n b_i X_i, c + \sum_{j=1}^m d_j Y_j) = \sum_{i=1}^n \sum_{j=1}^m b_i d_j Cov(X_i, Y_j).$$

$$\bullet \text{ If } X_1, \dots, X_n \text{ are uncorrelated, } Var(\sum_{i=1}^n X_i) = \sum_{i=1}^n Var(X_i).$$

$$\bullet \text{ } E[(X - \theta)^2] = Var(X) + (\mu_X - \theta)^2$$

(Mean square error = variance + bias square).

Common Mistakes in Using Expectation

① $\underline{E[g(X)h(Y)]} \neq E(g(X))E(h(Y))$. true when independent X & Y .

② $\underline{E[g(X)]} \neq g(E(X))$. when $g(x) = a + bx$
 $E(g(X)) = a + bE(X) = g(E(X))$.

$\int_{-\infty}^{\infty} e^x f(x) dx$
 $\underline{E(e^x)} \neq e^{E(x)}$

③ If X, Y are independent, then $\underline{\text{Cov}(X, Y)} = 0$, i.e., X, Y are uncorrelated. **The converse statement is true.**

X, Y independent, $\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$
 $= E(X)E(Y) - E(X)E(Y) = 0$.

But, if $\text{Cov}(X, Y) = 0$, $\Rightarrow X$ & Y are uncorrelated,

$\nRightarrow X$ & Y are independent $f(x, y) \neq f_X(x) \cdot f_Y(y)$

pdf $f(x)$

Moment Generating Function

- The **moment generating function (mgf)** of X is r.v. X , Let $Y = e^{tX}$, r.v.

$$\frac{d^2 M_X(t)}{dt^2} \Big|_{t=0} = E(e^{tX} \cdot X^2) \Big|_{t=0} = E(X^2)$$

$$M_X(t) = E(e^{tX}), \quad t \in \mathbb{R}$$

$$\frac{dM_X(t)}{dt} \Big|_{t=0} = E(e^{tX} \cdot X) \Big|_{t=0} = E(X)$$

if the expectation exists.

- The mgf may or may not exist for any particular value of t .
- Uniqueness theorem:** If the mgf exists for t in an open interval containing zero, it uniquely determines the probability distribution.
- If the mgf exists in an open interval containing zero, then $M_X^{(k)}(0) = E(X^k)$.
- For any constants a, b , $M_{a+bX}(t) = e^{at} M_X(bt)$.
- If X, Y independent then $M_{X+Y}(t) = M_X(t) M_Y(t)$.

$$E(e^{t(X+Y)}) = E(e^{tX} \cdot e^{tY}) = E(e^{tX}) \cdot E(e^{tY})$$

Joint Moment Generating Function

- For random variables $X_1, \dots, X_n$, their **joint mgf** is denfined as:

$$M_{X_1, \dots, X_n}(t_1, \dots, t_n) = E(e^{t_1 X_1 + \dots + t_n X_n})$$

if the expectation exists.

- 1 $M_{X_1}(t_1) = M_{X_1, \dots, X_n}(t_1, 0, \dots, 0)$.
- 2 **Uniqueness theorem.**
- 3 $X_1, \dots, X_n$ are independent if and only if

$$M_{X_1, \dots, X_n}(t_1, \dots, t_n) = \prod_{i=1}^n M_{X_i}(t_i).$$

- 4 $\frac{\partial^{r_1 + \dots + r_n}}{\partial t_1^{r_1} \dots \partial t_n^{r_n}} M_{X_1, \dots, X_n}(0, \dots, 0) = E(X_1^{r_1}, \dots, X_n^{r_n})$

Application

Exercise 5 (Portfolio returns):

R_A = return on asset A with $E(R_A) = \mu_A$ and $Var(R_A) = \sigma_A^2$,
 R_B = return on asset B with $E(R_B) = \mu_B$ and $Var(R_B) = \sigma_B^2$,
 $Cov(R_A, R_B) = \sigma_{AB}$.

Portfolio: x_A = share of wealth invested in asset A,

x_B = share of wealth invested in asset B,

and $x_A + x_B = 1$ (exhaust all wealth in 2 assets).

Then Portfolio return $R_P = x_A \cdot R_A + x_B \cdot R_B$.

$$\max_{x_A} (E R_P) \\ E R_P = x_A E R_A + x_B E R_B$$

Portfolio problem: how much wealth should be invested in assets A and B?

Please find portfolio expected return (gain from investing) and portfolio variance and standard deviation (risk from investing).

Conditional Expectation

$$E(Y) = \text{const.}$$

not random.

$$E(Y|x^*) = g(x^*)$$

Let x^* varies randomly, $E(Y|X) = \underline{g(X)}$
r.v.

$$E(E(Y|X))$$

- The **conditional expectation** of $h(Y)$ given $X = x$ is

Discrete case: $E(h(Y)|X = x) = \sum_y h(y) \underline{p_{Y|X}(y|x)}$

In particular,

$$E(\underline{Y|X=x}) = \sum_y y p_{Y|X}(y|x)$$

Continuous case: $E(h(Y)|X = x) = \int h(y) \underline{f_{Y|X}(y|x)} dy$

In particular,

$$E(Y|X=x) = \int y f_{Y|X}(y|x) dy$$

$f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx$
let $g(y) = f(x^*, y)$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

Question: 1) For joint pdf $f(x,y)$, if fix x^* , is $\underline{f(x^*, y)}$ a pdf of y ? **No.**

$$\int_{-\infty}^{\infty} f(x^*, y) dy \neq 1$$

2) $\underline{f_{Y|X}(y|x^*)} = f(x^*, y) / f_X(x^*)$ is a pdf of y . Why?

• $E(\underline{Y|x^*})$: the mean of $\underline{f_{Y|X}(y|x^*)}$. Do it for any $x = x^*$, and get a function of $x \Rightarrow E(Y|x)$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

Let $x = x^*$

$$f_{Y|X}(y|x^*) = \frac{f(x^*, y)}{f_X(x^*)}$$

$$\int_{-\infty}^{\infty} dy \text{ on both sides.}$$

$$\checkmark \int_{-\infty}^{\infty} f(x^*, y) dy = 1$$

Illustration

Example. $X = \text{age}$ (in year), $Y = \text{height}$ (in cm).

- $\underline{Y|X = x}$: r.v.
- $\underline{E(Y|X = x)}$: not a r.v. *const. $E(Y|x)$*
- $\underline{\text{Var}(Y|X = x)}$ and $\underline{\text{Var}(Y|X)}$ can be similarly defined.
- $\underline{E(Y)}$:
- $\underline{\text{Var}(Y)}$:

$E(Y|x) \Rightarrow \text{r.v.}$

Properties of conditional expectation

- ① $E_{Y|X}(h(Y)|x)$ is a function of x and is free of Y .
- ② If X and Y are independent then $E_{Y|X}(h(Y)|x) = E_Y(h(Y))$.
- ③ $E(h(X)|X = x) = h(x)$.
- ④ Let $g(x) = E_{Y|X}(h(Y)|x)$, then $g(X)$ is a random variable and denoted by $E_{Y|X}(h(Y)|X)$.
- ⑤ **Law of total expectation:** $E_X[E_{Y|X}(h(Y)|X)] = E_Y[h(Y)]$. In particular, $E_X[E_{Y|X}(Y|X)] = E_Y(Y)$.

- ⑥ **Variance decomposition:**

$$\text{Var}_Y(Y) = \text{Var}_X[E_{Y|X}(Y|X)] + E_X[\text{Var}_{Y|X}(Y|X)].$$

Note. 1. $\text{Var}_Y(Y) \geq \text{Var}_X[E_{Y|X}(Y|X)]$ and the equality holds if and only if $E_{Y|X}(Y|X) = E_Y(Y)$ with probability one.

2. $\text{Var}_Y(Y) \geq \text{Var}_X[E_{Y|X}(Y|X)]$ and the equality holds if and only if $\text{Var}_{Y|X}(Y|X) = 0$ with prob one, i.e., $Y = E_{Y|X}(Y|X)$ with prob one.

*Supplementary example

$$E(Y|X) = \underbrace{a + bX}_{g(X)} \quad \checkmark \quad (X, Y) \sim N_2(\mu, \Sigma)$$

Example 1.16. Predicting the value of a r.v. Y from another r.v. X .

- Data: X and Y (e.g. log return on a market index and log return on some stock of your interest)
- Statistical modeling: assign (X, Y) a known joint distribution (cdf, pdf or pmf)
- Objective: Predict Y using a function of X , i.e., $g(X)$.

We consider the following 3 groups of g 's:

- (i) $G_1 = \{g(x) : g(x) = c, \text{ where } c \in \mathbb{R}\}$
- (ii) $G_2 = \{g(x) : g(x) = a + bx, \text{ where } a, b \in \mathbb{R}\}$
- (iii) $G_3 = \{g(x) : g \text{ is arbitrary}\}$

Note. $G_1 \subset G_2 \subset G_3$.

- Question: Within each group, what is the “best” prediction?
- Criterion: minimizing mean square error:

$$MSE = E_{XY}[(Y - g(X))^2]$$

- **Solution (i).** “Best” constant prediction:
Since

$$\begin{aligned} E_{XY}[(Y - c))^2] &= E_Y[(Y - c))^2] &= Var(Y - c) + [E_Y[Y - c]]^2 \\ &= Var_Y(Y) + (\mu_Y - c)^2 \\ &\geq Var_Y(Y) \equiv E_Y[(Y - \mu_Y))^2] \end{aligned}$$

The equality holds iff $c = \mu_Y$.

- **Solution (ii).** “Best” linear prediction:

Since

$$\begin{aligned}
 E_{XY}[(Y - a - bX)^2] &= \text{Var}(Y - a - bX) + [E_{XY}(Y - a - bX)]^2 \\
 &= \text{Var}_{XY}(Y - bX) + (\mu_Y - a - b\mu_X)^2 \\
 &\geq \text{Var}_{XY}(Y - bX) \quad (\Rightarrow \text{setting } a = \mu_Y - b\mu_X) \\
 &= \sigma_Y^2 + b^2\sigma_X^2 - 2b\sigma_{XY} = \sigma_X^2 \left(b^2 - 2b\frac{\sigma_{XY}}{\sigma_X^2} + \frac{\sigma_{XY}^2}{\sigma_X^4} \right) + \sigma_Y^2 - \frac{\sigma_{XY}^2}{\sigma_X^2} \\
 &= \sigma_X^2 \left(b - \frac{\sigma_{XY}}{\sigma_X^2} \right)^2 + \sigma_Y^2(1 - \rho^2) \\
 &\geq \sigma_Y^2(1 - \rho^2) \quad (\Rightarrow \text{setting } b = \frac{\sigma_{XY}}{\sigma_X^2} = \rho \frac{\sigma_Y}{\sigma_X})
 \end{aligned}$$

error

The equality holds iff $a = \mu_Y - b\mu_X$ and $b = \rho \frac{\sigma_Y}{\sigma_X}$. The “best” linear predictor is $\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X)$.

- **Solution (iii).** “Best” prediction of Y using X :
Since

$$\begin{aligned}
 E_{XY}[(Y - g(X))^2] &= E_{XY}[Y - E_{Y|X}(Y|X) + E_{Y|X}(Y|X) - g(X)]^2 \\
 &= E_{XY}[Y - E_{Y|X}(Y|X)]^2 + E_X[E_{Y|X}(Y|X) - g(X)]^2 \\
 &\quad + 2 \cdot E_{XY}\{[Y - E_{Y|X}(Y|X)][E_{Y|X}(Y|X) - g(X)]\} \\
 &= E_{XY}[Y - E_{Y|X}(Y|X)]^2 + E_X[E_{Y|X}(Y|X) - g(X)]^2 \\
 &\geq E_{XY}[Y - E_{Y|X}(Y|X)]^2
 \end{aligned}$$

The equality holds iff $g(X) = E_{Y|X}(Y|X)$.

Note:

- $E_{Y|X}(Y|X)$ is the best predictor of Y based on X in the mean square prediction error sense.
- $E_{Y|X}(Y|X)$ is called the **regression function of Y on X** .

Also note that we need to know the joint distribution of X and Y , or at least $E_{Y|X}(Y|x)$ to get the “best” prediction of Y using X .

- $\min_g E(Y - g(X))^2 = \min_{a,b} E(Y - a - bX)^2$ iff

$$E_{Y|X}(Y|x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X).$$

- It is true when (X, Y) is distributed as bivariate normal.
- Equivalently, need to know the means, variances and covariances only.
- The prediction error $\sigma_Y^2(1 - \rho^2)$ is small if ρ is close to ± 1 .

$$E(g(X)) \neq g(E(X)).$$

δ Method *if $g(\cdot)$ is nonlinear*

Let $Y = g(X)$. Suppose we know the mean μ_X and variance σ_X^2 only, but **not** the entire distribution.

Question 1: Can we derive the distribution of Y ?

Question 2: Can we “roughly” describe the mean and variance of Y ?
 $(E[g(X)] = g[E(X)]?)$

• δ Method:

$$\begin{aligned}
 Y = g(X) &\approx g(\mu_X) + (X - \mu_X)g'(\mu_X) \quad (\text{by Taylor expansion}) \\
 \Rightarrow E[g(X)] &\approx g(\mu_X) = g(E(X)) \\
 \text{Var}[g(X)] &\approx \text{Var}(X)[g'(\mu_X)]^2 \\
 &\approx \text{Var}(g(\mu_X)) + g'(\mu_X)^2 \cdot \text{Var}(X - \mu_X) = 0 + g'(\mu_X)^2 \cdot \text{Var}(X) \\
 \text{or } Y = g(X) &\approx g(\mu_X) + (X - \mu_X)g'(\mu_X) + \frac{1}{2}(X - \mu_X)^2 g''(\mu_X)
 \end{aligned}$$

when $X \approx \mu_X$ ✓

Note. How good these approximations are depends on whether g can be reasonably well approximated by the 1st or 2nd order polynomials in a neighborhood of μ_X and on the size of σ_X .